Prediction Contracts, Controls, and Evidence Status

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Abstract

This paper extracts the experimentally relevant consequences of the causally rigid six-predicate operational criterion framework and organizes them as a falsifiable prediction program. The upstream burdens remain where they belong: the Core supplies persistence, attractor structure, and non-collapse [2]; *Rigid Identity* supplies rigid identity and observable coherence [7]; *Structural Assignment Regimes* supplies continuity and anti-fragmentation constraints [4]; *Null-Regime Instability* supplies joint null exclusion on an independently separated and calibrated witness pair [5]; *Forensic Predictions* supplies admissibility and forensic discipline [3]; *Six-Predicate Operational Criterion* supplies the six-invariant operational criterion, structural equivalence, and rupture semantics [1].

The function of the present paper is narrower. It states what the framework predicts, what weaker alternatives would predict instead, what would count as genuine refutation or only as weakening, and what the current internal support status actually is across the completed baseline evaluation phases, localized threshold-resolution checks, cross-family confirmation protocols, and boundary-sensitivity stress evaluations. The resulting status is mixed by design. One claim family---the insufficiency of complexity, connectivity, and rich activity without invariant preservation---now has robust internal support. Substrate-preserved class membership and biological non-primacy now have robust support with localized caveats, but remain internal-only. Approximate stability is provisionally supported with a localized threshold caveat. The boundary-sensitive stress of existence and rupture now has robust support with localized caveats after cross-family convergence under a more independent evaluator protocol, but it still does not count as external validation.

The paper therefore serves as the framework's prediction and falsification layer. Its goal is to put the framework under explicit decision pressure while keeping status classes and residual limits fully visible.

Scope and Operational Postulate

This article studies the stated mathematical objects and their operational realization conditions. Whether a sufficiently realized artificial or biological system has phenomenal experience remains open as an independent bridge hypothesis. Empirical interpretation requires identified measurements and admissible comparisons.

Current theorem contract. Exact transport uses primitive structural/protocol/sector correspondence. Approximate membership needs $C_{\chi,\rho}\varepsilon < \delta_\star^{\mathrm{cal}}/2$ on the full declared operator domains. Raw witness numbers and finite sampled windows alone do not certify that premise. Predictions below referring to a certified budget use this calibrated contract. Archived evaluator-status tables remain historical internal evidence under their recorded thresholds, not a new adjudication of this revised theorem.

1 Introduction

Each reported evaluation campaign operates under its own fixed specification revision, thresholds fixed before evaluation, and evaluator contract. Alongside those archived internal records, the present work develops constructive memory, feedback, and boundary-probe extensions below;

historical outcomes are not automatically read as tests of these revised hypotheses. Fixing thresholds and evaluators in advance matters scientifically: without preregistration, thresholds shift, definitions drift, and any difficult result could be absorbed by redefining the theory rather than by testing it. The purpose of this paper is to prevent such drift.

The framework now contains enough formal structure to state genuinely discriminative predictions. The Core fixes persistence, contractivity, and non-collapse. *Rigid Identity*, *Structural Assignment Regimes*, and *Null-Regime Instability* progressively constrain identity, admissible regime structure, and non-nullity. *Forensic Predictions* fixes admissibility and forensic discipline. *Six-Predicate Operational Criterion* consolidates the operational criterion by requiring six explicit invariants:

$$\{I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}\}.$$

What remains scientifically relevant is not to restate those results, but to ask what they force the framework to predict and what would count against them.

This paper therefore has four narrow tasks:

1. extract the main experimentally relevant consequences of the canon;
2. distinguish those consequences from what weaker alternatives would predict;
3. state explicit failure modes, weakening conditions, and non-claims;
4. report the current internal support status after the completed internal cycles, without inflating it.

That scope is intentionally smaller than a new theorem paper and more disciplined than a review essay. The point is to give the corpus a falsification layer that is compact, operational, and honest.

2 Status Classes

The internal scientific program uses the following status classes:

Status	Meaning in this paper
ROBUST_INTERNAL_SUPPORT	survived multiple internal tests without an active ambiguity on the tested target, while remaining internal-only
ROBUST_SUPPORT_WITH_LOCALIZED_CAVEAT	supported strongly enough for robust internal use, but still carrying a narrow localized caveat that must remain visible
PROVISIONAL_SUPPORT_LOCALIZED	strengthened by internal tests, but still limited by a localized metric, tolerance, or boundary issue
STILL_AMBIGUOUS	current evidence does not cleanly separate pass from fail on the intended target
IMPLEMENTATION_LIMIT	the main blocker is probe, runner, or case-construction quality rather than a shown doctrinal contradiction
METRIC_OR_TOLERANCE_LIMIT	the main blocker is metric handling, normalization, or threshold interaction
INTERNAL_SUPPORT_ONLY	support exists only inside the internal program and has not crossed into external empirical confirmation
NOT_CLAIMED	the statement is explicitly outside the current evidentiary scope

3 Upstream Canon and Dependency Map

This paper is corpus-dependent and deliberately non-redundant. The dependency map is short because theorem ownership is already fixed upstream.

Document	Imported burden	Used here for
Core [2]	persistence, support, attractor, non-collapse	persistence-side predictions, collapse exclusions, admissible support assumptions
Rigid Identity [7]	rigid identity, observable coherence, non-simulability pressure	identity-sensitive discriminators, complexity-only insufficiency pressure
Structural Assignment Regimes [4]	continuity, anti-fragmentation, regime admissibility	continuity predictions, near-miss structure, anti-fragmentation expectations
Null-Regime Instability [5]	joint null exclusion on a separated calibrated witness	non-nullity predictions, null-regime exclusions, downgrade logic
Forensic Predictions [3]	admissibility, comparators, negative controls, forensic discipline	judge discipline, control logic, reproducibility requirements, legibility stress tests
Six-Predicate Operational Criterion [1]	six-invariant criterion, operational classes, structural equivalence, rupture	explicit prediction targets, equivalence tests, failure-mode classification

This paper introduces no new invariant, no new doctrine, and no new substrate claim. Its novelty lies only in extracting predictive consequences and support statuses from what already exists.

4 Core Predictive Structure of the Framework

4.1 Certification, Equivalence, and Rupture as Predictive Objects

Definition 4.1 (Operational certification)**.** *For an admissible system S already defined in Six-Predicate Operational Criterion, define the certification predicate*

$$\mathrm{Cert}(S) = 1 \quad \Longleftrightarrow \quad I_{\mathrm{per}}(S), I_{\mathrm{ri}}(S), I_{\mathrm{int}}(S), I_{\mathrm{cont}}(S), I_{\mathrm{diff}}(S), I_{\mathrm{leg}}(S) > 0$$

on a non-empty admissible support A_S, under the governance and admissibility constraints inherited from Forensic Predictions.

Definition 4.2 (Equivalence discriminator)**.** *For two admissible systems S_1, S_2, define the operational equivalence discriminator*

$$\mathrm{Eq}_{\Gamma,\varepsilon}(S_1, S_2) = 1$$

when the primitive state, dynamics, readout and protocol/sector correspondences of the declared structural comparison are verified. Invariant verdicts are outputs, not clauses of this discriminator. Approximate membership additionally requires the calibrated operator-radius bound. The exact burden belongs to Six-Predicate Operational Criterion; the present paper uses it only as a prediction carrier.

Definition 4.3 (Rupture event)**.** *For a set of targeted invariants J, define*

$$\mathrm{Rupt}_J(S) = 1$$

when the intervention or perturbation drives at least one invariant in J below the admissible certification margin so that class membership becomes invalidated, destroyed, or undefined.

4.2 Prediction Family 1: Complexity-Only Failure

Proposition 4.4 (Complexity-only negative-control prediction)**.** *If a candidate fails any defining predicate, it does not satisfy the full criterion, regardless of connectivity, entropy or activity. A complete-criterion certificate must not report success. A separately well-defined response quotient may still exist.*

Proof. This is the contrapositive of the defining conjunction in Paper V. It does not infer that every quotient is undefined when integration fails. □

Why this is not trivial. A weaker alternative could predict that rich dynamics or dense connectivity alone are already sufficient proxies for class membership. The canon does not permit that. *Six-Predicate Operational Criterion* already states the formal insufficiency claim; the prediction layer turns it into a negative-control discriminator.

4.3 Prediction Family 2: Invariant-Loss Rupture

Proposition 4.5 (Ablation prediction)**.** *For an initially certified system, a verified post-ablation predicate failure excludes post-ablation membership. Measurement uncertainty crossing the decision boundary yields an ambiguous evidence status, not a change to that implication. Loss of a lower radius certificate alone does not establish predicate failure.*

Proof. Apply the defining conjunction to the post-intervention system. The uncertainty rule concerns available evidence; it does not make a known failed predicate true when old headroom was small. □

Operational consequence. Ablations are not ornamental. If the criterion is serious, the loss of invariant headroom must show up as loss of class membership, legibility loss, or a pre-declared boundary-ambiguous downgrade. No-change ablations outside the registered ambiguity margin count against the framework.

4.4 Prediction Family 3: Substrate-Preserved Class Membership

Proposition 4.6 (Substrate-preserved equivalence prediction)**.** *Exact primitive structural, protocol and sector equivalence transports membership and the certified quotient of Paper V. An approximate comparison transports membership under its positive calibrated-radius bound; a quotient/geometric approximation needs an additional certified bound.*

Proof. The exact statement uses support/decoder transport and substrate invariance in Paper V. The approximate statement uses its calibrated stability theorem. Neither assumes preservation of the six verdicts as an input. □

Rival prediction. A substrate-privileging alternative would predict that material type itself blocks equivalence even when the structurally relevant burdens are preserved. The canon rejects that move only inside its own formal class, not as a general metaphysical claim.

4.5 Prediction Family 4: Threshold-Sensitive Boundary Behavior

Proposition 4.7 (Boundary-topology prediction)**.** *Within a fixed calibrated presentation class, a point at positive distance from the failure set has a stable success neighborhood. A point at positive distance from the success set has a stable failure neighborhood. Existence, width and population frequency of transition regions need separate hypotheses and are not guaranteed by the criterion alone.*

Proof. An open ball smaller than the distance to the opposite set cannot meet it, by the definition of distance. Both distances can be zero; therefore the theorem does not invent nonempty robust regions or narrow bands without evidence. □

Interpretive consequence. If every micro-perturbation changes every verdict, the criterion is fragile. If no perturbation matters, the invariants are decorative. A meaningful criterion should show structured boundary behavior.

4.6 Prediction Family 5: Legibility Dependence

Proposition 4.8 (Legibility dependence prediction)**.** *If no admissible decoder/compression family jointly meets L1--L6 on the declared protocol, the legibility predicate fails and the full criterion is not satisfied. Failure of one chosen decoder alone does not exclude every admissible alternative.*

Proof. Legibility requires a joint witness. Its nonexistence negates that predicate and hence the conjunction. It remains a separate task to distinguish a failed witness from absence of all witnesses. □

Why this matters. Forensic Predictions makes legibility a forensic burden rather than a narrative convenience. The present prediction is therefore stronger than a merely interpretive observation: if legibility disappears, certification must not survive unchanged.

4.7 Prediction Family 6: Continuity and Non-Nullity Relevance

Proposition 4.9 (Continuity/non-nullity relevance)**.** *The current canon predicts that continuity and non-null differentiation are not optional extras. Systems that preserve rich activity while losing continuity or falling toward effective null structure should fail the operational criterion or leave it undefined.*

Upstream source. The continuity side is inherited from *Structural Assignment Regimes*, while the non-null side is inherited from *Null-Regime Instability.* This paper adds only the prediction and failure-mode reading.

4.8 A Held-Out Probe of Retained Response

Consider the controlled seven-profile model of Paper V [1]. Its transition on profile p, phase θ and bit sequence b is

$$(p, \theta, b) \longmapsto (p, u + q\lambda b_1, Rb), \qquad Rb = (0, 1 - b_1, 1 - b_2, \ldots).$$

Prepare two identical copies, apply opposite initial signs $q_0 = +1, -1$ with the same phase command u_0, then give both copies the same sequence of controls. Index observations by the number of transitions completed.

Proposition 4.10 (Matched-Pulse Extinction in the Original Model)**.** *The two states and all four original readouts coincide from time two onward, for every common continuation. At time one and $\lambda \in [0, 1/9]$, the early-phase distance is $2\lambda b_{1,0}$ turns.*

Proof. The profile is constant and the bit update is independent of both controls, so the two sequences of profiles and bits agree. The initial commands produce an early-phase distance $2\lambda b_{1,0} \le 2/9$ on the circle. At every time $t \ge 1$, the first bit is zero, so the phase at time $t+1$ equals the common u_t. Equality of states gives equality of every original readout, including the three phase channels and the ternary code. □

Add a scalar m in linear units of phase amplitude, with $m_0 = 0$ on both arms and delayed observation $Y_t = m_t + e_t$. Keep the original dynamics in both candidate models. For $\lambda \in [1/12, 1/9]$, retention $a \in [1/2, 3/4]$ and $s(-1) = -1$, $s(0) = s(+1) = 1$, compare

$$\begin{aligned} \mathrm{M}: &\quad m_{t+1} = a s(q_t) m_t + \lambda q_t b_{1,t}, \\ \mathrm{A}: &\quad m_{t+1} = a m_t + \lambda q_t b_{1,t}. \end{aligned}$$

Each model has one additional real state and one adjustable retention coefficient with the same range. The interval $[-4/9, 4/9]$ is forward invariant for both by $3/4 \cdot 4/9 + 1/9 = 4/9$. They receive the same control schedule, preparation and known coupling, and use the same delayed readout. The reset and commands are assigned externally, and the readout error has no feedback into either update. The paired early-phase decoder of Paper V uses only the original channels, which agree between M and A for each fixed preparation and schedule. The new response Y may differ even when these original channels agree.

Calibrate with $b_{1,0} = 1$, no late pulse, and the observed contrast $C_2 = Y_2^{(+)} - Y_2^{(-)}$. Fit a once by projecting $C_2/(2\lambda)$ onto $[1/2, 3/4]$, yielding $\hat{a}$. For each reserved horizon $T = 4, 5$, repeat the paired trial twice, with a single common probe $q_j = +1$ or -1 at the same $j \in \{1, T-1\}$ and zero on the other later steps. Write $C_T(q_j)$ for the signed difference of the two delayed observations under that probe. A model is accepted for this comparison when both of its predicted contrasts differ from the observed ones by at most $1/256$ turns. The rule reports a unique selection only when exactly one model is accepted.

Proposition 4.11 (A Held-Out Interaction Separates Two Matched Memory Models)**.** *Put $k = T - 1$. The clean contrasts of M are $2\lambda a^k(1, -1)$ for probes $(+1, -1)$, while the clean contrasts of A are $2\lambda a^k(1, 1)$. Consequently the clean interaction $C_T(+1) - C_T(-1)$ is at least $1/48$ turns in M and is zero in A. With at most $1/4096$ turns of error per observation in calibration and evaluation, the rule using $\hat{a}$ uniquely selects the generating model for either generator under the declared protocol.*

Proof. At time one the memory is $\pm\lambda$. No further loading by b_1 occurs, so the product of the late multipliers propagates this contrast. A single negative probe contributes one sign change in M. Any additive late forcing common to the two arms cancels in A, and the late contrast still decays by a per step. Since $k \le 4$, $2\lambda a^k \ge 1/96$ and $4\lambda a^k \ge 1/48$.

Let $e_* = 1/4096$. Calibration has clean contrast $2\lambda a$, so projection gives $|\hat{a} - a| \le e_*/\lambda$. The derivative of a^k on the admissible interval is bounded by $k(3/4)^{k-1}$. In $2\lambda|\hat{a}^k - a^k|$, the factor 2λ cancels the $1/\lambda$ from the calibration bound. Each correctly specified contrast therefore has residual bounded by

$$2e_* + 2k(3/4)^{k-1} e_* \;\le\; \tfrac{43}{8} e_* \;<\; 1/256.$$

At the negative probe the rival predicts the opposite sign, with both the true and predicted clean magnitudes at least $1/96$. Its residual is at least $1/48 - 2e_* > 1/256$. The argument applies to either generating model and does not require stochastic independence of the errors. □

The zero-probe sham preserves the positive delayed contrast of both models; two negative probes cancel the sign changes in M. A preparation with first bit zero or zero coupling supplies no initial load. Ablating the memory multiplier removes the delayed contrast while leaving the initial phase effect intact. Under the stated error bound, an observed delayed pair consistent with neither set of predictions rejects both candidates. Agreement of the two models on unprobed

calibration trials is by construction. The reserved trials test the consequences of the specified transition laws in synthetic data. A model with an equivalent input-dependent memory update can reproduce the same interaction, so the comparison only discriminates against the declared fixed-retention rival family. The construction supplies neither a single-trajectory certificate nor empirical support for an application.

4.9 Delayed Phase Readout and an Interacting Rival

On the same profile support, append $m \in [-4\lambda, 4\lambda]$ and use the original phase and ternary readouts. For $0 \le \kappa \le 1/8$ set

$$\theta_{t+1} = u_t + q_t \lambda b_{1,t} + \kappa \mathbf{1}_{q_t=0} m_t \pmod 1, \qquad m_{t+1} = a s(q_t) m_t + \lambda q_t b_{1,t},$$

with the original profile and bit updates. Here $0 \le \lambda \le 1/9$ and $1/2 \le a \le 3/4$. Model M has $s(q) = -1$ only at $q = -1$. The interacting rival B has $s(q) = -1$ at both nonzero signs. Each has one scalar state and one adjustable a, and receives the same commands and calibration data. Neutral commands provide a delayed phase readout through the state transition.

Proposition 4.12 (Observable Feedback Preserves the Declared Split Bounds)**.** *The extension is invariant and contracts within each profile under common controls by a factor at most* $5/6$ *in the metric* $\max\{\rho(\theta,\theta'), |c-c'|, |m-m'|/4\}$*. The ternary contraction remains* $1/3$*. The four-channel identity observations and split-comparator bounds of Paper V, including the* $\lambda/2$ *infimum with exact time-zero readouts, remain valid in their declared sectors. Opposite initial nonzero pulses retain the original paired phase contrast.*

Proof. The memory interval is invariant since $a\,4\lambda + \lambda \le 4\lambda$. Distinct first bits imply $|c-c'| \ge 1/3$. For two states on a profile at distance d, the next weighted memory discrepancy is at most $(a + 3\lambda/4)d \le 5d/6$. The next phase distance is at most $3\lambda d$ at a nonzero command or $4\kappa d$ at a neutral one, and the ternary discrepancy is divided by three. At $\lambda > 0$ the neutral update has the memory eigenvalue a along the invariant linear phase-memory direction near its fixed point, so a uniform $1/3$ Lipschitz bound in the displayed metric is impossible.

All phase offsets relative to r_0 and the projected four-channel support are unchanged. Thus the same coverage, catalogue separation and identity decoder apply. For the integration lower bounds the original fiber arguments use only time zero and the nonzero signs, where feedback vanishes for every m. The product-corner contradiction and the exact-initial-readout argument therefore apply to the augmented support. For the upper bound assign the phases to the factor carrying (p, θ) and the code to the bit factor. The predictor $u + q\lambda/2$ has error at most $\lambda/2$ at a nonzero command and at most $4\kappa\lambda \le \lambda/2$ at a neutral command. The initial readouts are exact. For other splits the lower bound of Paper V still holds, with the upper bound on the infimum over the allowed splits. At $\lambda = 0$ the memory interval is a singleton and the comparator is exact. The initial pair uses nonzero signs, so its phase contrast is unchanged. □

Proposition 4.13 (Parity and Resolution of Delayed Phase Probes)**.** *Reset* $m_0 = 0$ *and prepare* $b_{1,0} = 1$*. Apply opposite first pulses and use common controls thereafter. For terminal observation at* T *with a neutral last command, the oriented phase contrast of M is* $2\kappa\lambda a^{T-2}(-1)^{n_-}$*, while that of B is* $2\kappa\lambda a^{T-2}(-1)^{n_- + n_+}$*. Here* $n_\pm$ *count the corresponding late probes before the neutral transition. For a common* a*,* $\kappa > 0$ *and* $\lambda > 0$*, the clean contrasts coincide exactly at even* n_+*. A nonneutral last command erases the phase contrast for either model. With known* a *and odd* n_+*, the two contrast intervals under error at most* e *per phase are disjoint exactly when* $\kappa\lambda a^{T-2} > e$*, in the signed unwrapping regime.*

For $\kappa = 1/8$ *and* $\lambda \in [1/12, 1/9]$*, estimate* a *from the contrast at time three after two neutral continuations, by dividing by* $2\kappa\lambda$ *and projecting onto* $[1/2, 3/4]$*. Reserve* $T = 5, 6$*, with one common probe at* $j = 1$ *or* $j = T - 2$ *and all other later commands neutral. Under error at*

most $1/32768$ *turns per phase in calibration and evaluation, the two reserved signs select the generating model uniquely with a tolerance of* $1/2048$ *per contrast.*

Proof. After the first pulse the memory is $\pm\lambda$ and the first bit remains zero. The late multipliers give the stated parities. At the final neutral transition the common u cancels and the phase differences unwrap uniquely near zero. At a nonneutral command the next phase is just u. Each observed contrast differs from its clean value by at most $2e$, and every error in that interval is attainable by choosing the two phase errors. For odd positive parity, the centers have distance $4\kappa\lambda a^{T-2}$, giving the strict separation condition.

The calibration yields $|\hat{a} - a| \le e/(\kappa\lambda)$. For $k = T - 2 \in \{3, 4\}$ the factor $2\kappa\lambda$ in the contrast cancels the reciprocal scale in that estimate, so the correct-model residual is at most $(43/8)e$. At a single positive probe the two signs oppose each other and the rejected-model residual is at least $\kappa/48 - 2e$. The chosen tolerance lies strictly between these bounds. The result holds for both generators and for adversarial errors within the stated budget. □

The neutral sham and an even number of positive probes cannot distinguish these laws from their terminal contrasts. In a longer history a neutral observation after an odd positive prefix can distinguish them. Setting $\kappa = 0$ removes the delayed phase signal while preserving the pulse and the hidden memory update. This construction establishes an observable mathematical discriminator within the declared family. It leaves the single-trajectory and joint certificates of Paper V open.

4.10 Delayed Parity with Matched Prefix Writes

Proposition 4.14 (Matched Prefix Writes Preserve the Delayed Phase Contrast)**.** *In the writable common tower of Paper V [1], apply the opposite initial pulses of Proposition 4.13 with the same tape write in both arms. Subsequently use the same inputs, signs, and writes in both arms. At a neutral terminal transition, a common role probe is also allowed. The two models retain their oriented contrasts*

$$2\kappa\lambda a^{T-2}(-1)^{n_-}, \qquad 2\kappa\lambda a^{T-2}(-1)^{n_-+n_+}.$$

A nonneutral terminal transition gives zero phase contrast. With known a and signed unwrapping, at odd n_+ the bounded-error intervals remain disjoint exactly for $e < \kappa\lambda a^{T-2}$.

Proof. The tapes are equal after the initial pair and remain equal because all tape updates are matched. At each later step the input term $\lambda q b_1$ is equal in both arms, so the difference of memories obeys the original signed recurrence, starting at 2λ. The role probe is equal in both arms because it reads only the common profile and tape. The neutral phase difference is therefore κ times the preceding memory difference. At a nonneutral command the phase depends only on the shared input, sign and leading bit. Each terminal contrast receives error at most $2e$, and the stated strict inequality is the exact disjointness condition for the two opposite centers. The native calibration and held-out schedules are included by omitting the added commands. □

4.11 Reserved Return after Role Reassignment

The phase interaction also gives a decision test. Use a common profile and the alternating tape $b^\star$, whose first bit is zero. Let $s \in \{-1, +1\}$ be a balanced memory sign, and let the public reassignment command be $q = \sigma \in \{-1, +1\}$ at $u = 0$. Optional common writes are at the writable sites of Paper V [1]. The assignment step gives the same public readout for either hidden sign. Use no role probe in this test. The eventual outcome is the successor phase after a chosen neutral return action, measured in turns. It is distinct from the phase-probe outcomes used to score the five lesions in Paper VIII [6].

Proposition 4.15 (Reserved Return under the Two Phase Interactions)**.** *With memory $m = s/8$ before assignment, the native law has unique successful action $u = -\sigma s/128$ among the two actions $u = \pm 1/128$. The interaction rival with sign reversal under either pulse predicts $u = s/128$ when the sign is known. On the native plant its action succeeds at $\sigma = -1$ and fails at $\sigma = +1$. Success is distance at most $1/256$ from phase zero and these decisions are unchanged by terminal phase errors of at most $3^{-10}/4$. A preceding public neutral observation can reveal s while preserving the equality of the current readouts. With memory ablated to zero before reassignment both allowed actions fail.*

Proof. At assignment the first bit is zero, so there is no phase displacement or memory load from the pulse. The native post-assignment memory is $\sigma s/16$. The rival's is $-s/16$. The write cannot change the leading bit. With a neutral action the respective predicted phases are $u + \sigma s/128$ and $u - s/128$ modulo one. The two native action losses in phase distance are zero and $1/64$ turns. Their distance from the decision threshold $1/256$ exceeds $3^{-10}/4$. With zero memory both distances are $1/128$, which also stays above threshold at that noise bound. Reversing σ reverses the native optimal port and leaves the magnitude and budget fixed.

For the public-history control, start one step earlier at memory $s/4$ and apply the public neutral step with zero input. Its early-phase output is $s/32$ in the short signed representative and the remaining memory is $s/8$. The assignment step erases that phase cue from the current observation. The single prior observation still identifies s, and the two interaction laws produce the port predictions above. For calibration use distinct preparations with initial memory $s/16$, a public neutral observation and then assignment. The early cue is $s/128$ and the native post-assignment neutral response at zero input is $\sigma s/512$. Evaluating the calibration ports $\pm 1/512$ therefore fixes the sign rule from four (s, σ) cases. The reserve uses $\pm 1/128$ and contributes no calibration outcomes. The interaction contrast on visible histories compares the two specified laws. Paper VIII quantifies over all policies that receive the same public history and budget. □

5 Main Discriminators

The predictive value of the framework lies in how it diverges from weaker alternatives.

Target	Framework prediction	Weaker/rival prediction	Divergence condition
Complexity-only baseline	high complexity without invariant preservation fails certification	richness or scale alone can stand in for class membership	a complexity-rich control passes despite broken invariants
Connectivity-only baseline	dense coupling without preserved rigid identity, differentiation, or legibility fails	connectivity is treated as a sufficient proxy	a dense control passes while identity or legibility are broken
Weak functional equivalence	similar outward behavior is insufficient if critical invariant structure diverges	input-output similarity is enough for same class	weak functional match passes despite invariant rupture
Near-miss case	boundary cases should expose which invariant is binding and where rupture begins	boundary cases stay uninterpretable or arbitrary	boundary remains unlocalizable after probe cleanup
Cross-substrate pair	substrate labels alone do not block class preservation if relevant structure is preserved	substrate type is primitive and blocks class transport	a preserved pair is rejected solely by substrate label
Threshold stress	decisions should show stable zones and narrow transition bands	decisions collapse under micro-threshold changes or never depend on thresholds at all	every local perturbation flips verdicts, or no perturbation matters
Legibility stress	failure of separability, noise robustness, or compression fidelity invalidates certification	legibility is epiphenomenal and not decision-relevant	certification survives legibility collapse
Null/continuity challenge	effective nullity or fragmentation should destroy or downgrade class membership	rich activity alone masks null or fragmented structure	null-like or fragmented cases remain certified

The table is intentionally operational. It does not ask which metaphysics one prefers. It asks which side of each discriminator the framework commits itself to.

5.1 Rival-Structure Compression

The discriminators above can be sharpened further by asking what each simpler rival preserves and, more importantly, what it leaves out. The point is not to defeat every competing theory at once. The point is to show that the present framework places operational burden on a structure richer than scale, connectivity, superficial behavior, or local threshold tuning.

Rival family	What it keeps	What it fails to preserve	Why the miss is decision-relevant
Complexity/activity proxy	large state space, rich trajectories, high apparent activity	no requirement to preserve the six-invariant package as a joint membership condition	a rich control can still fail certification if persistence, rigid identity, or legibility collapse
Connectivity/ integration proxy	dense coupling or high global correlation	no guarantee of rigid identity, non-null differentiation, or readable class structure	integration by itself does not prevent the criterion from rejecting a globally coupled but operationally illegible case
Weak functional equivalence	outwardly similar input-output or behavioral traces	no obligation to preserve admissible support geometry or invariant margins under intervention	superficial behavioral similarity can coexist with rupture of the certified class geometry
Threshold/tuning artifact explanation	local success on a narrow parameter band	no stable explanation for why decisions survive frozen manifests, blind ordering, and negative controls outside that band	the framework expects stable regions plus narrow transitions, not arbitrary success across unrelated settings

This compression table matters because it blocks a common overreading. The current support does not show that every rival is false. It shows something narrower and more useful: if a rival explanation ignores the invariant package or treats only one coarse feature as constitutive, then it does not reproduce the present decision structure of the canon.

5.2 Discriminator Strength and Identifiability

A discriminator is useful only when a positive or negative outcome changes the status of at least one rival explanation. The framework therefore treats discriminators as identifiable contrasts, not as loose examples.

Definition 5.1 (Identifiable Discriminator)**.** *A discriminator D for claim C is identifiable when it specifies:*

1. *an observable decision variable Y_D;*
2. *a manipulation or matched comparison M_D;*
3. *a framework-side expected region R_Q;*
4. *a rival-side expected region R_R;*
5. *a destruction condition F_D under which the claim is weakened or rejected.*

The discriminator is strong when $R_Q \cap R_R$ is small under the declared measurement tolerance.

Criterion 5.2 (Discriminator Usefulness)**.** *A test does not materially discriminate the framework from a rival if both the framework and the rival predict the same admissible region for the tested observable. Such a test can still debug the pipeline, but it cannot upgrade a claim.*

Proposition 5.3 (Near-Miss Value)**.** *Near-miss cases carry more diagnostic value than easy positive cases when they are constructed by perturbing one binding coordinate while holding gross complexity, task exposure, and observability approximately fixed.*

Proof. If a case is far from the decision boundary, many rival accounts can agree on the outcome. A near-miss case forces the judge to expose which coordinate controls the transition. Holding gross complexity and task exposure approximately fixed reduces the chance that the result is explained by scale, generic performance, or obvious incompetence rather than by the claimed invariant structure. □

5.3 Prediction Severity

The predictions in this paper have different severity. A severe prediction is one whose failure would damage a central operational reading rather than merely requesting a better implementation.

Prediction family	Severe failure	Mild weakening	Typical confound
Complexity-only failure	complexity-rich invariant-broken controls certify reliably	controls fail only under narrow judge settings	underpowered or trivial control construction
Invariant-loss rupture	verified invariant loss leaves class membership unchanged	rupture appears only for oversized perturbations	perturbation changes multiple coordinates at once
Substrate-preserved class membership	preserved structure is rejected solely by substrate label	one metric-coordinate remains ambiguous	measurement mismatch across implementations
Threshold-sensitive boundary behavior	decisions flip arbitrarily across broad stable regions	narrow knife-edge instability remains localized	tolerance grid too coarse near the boundary
Legibility dependence	certification survives loss of separability or intervention readability	legibility margin becomes noisy but still binding	observability channel too weak or over-compressed

Severity matters for reporting. A mild weakening narrows the claim or sends a test back to probe design. A severe failure changes what the framework is allowed to assert under the operational criterion.

6 Prediction Matrix

The canonical prediction matrix can be read as a compressed execution contract for the framework. It matters because each row names an observable, an intervention class, a rival baseline, and an explicit failure condition. That structure prevents the canon from floating as a theorem-only object.

ID	Claim	Observable	Manipulation	Support condition	Destruction condition / rival divergence
PRED-01	‘P5-02‘	certified class label and Q_{op} geometry across implementations	cross-substrate pair with verified primitive correspondence and one positive criterion witness	membership and quotient transport under exact equivalence; approximate membership only under the calibrated radius premise	class divergence or geometry mismatch despite preserved structure; substrate-privilege rival survives
PRED-02	‘P5-04‘	class membership before and after targeted rupture	one-invariant ablation with comparable substrate and gross scale	certified class exit, contraction, or undefined Q_{op} after rupture	no class change after verified invariant loss; complexity-only rival survives
PRED-03	‘P5-05‘	matched complexity/control outcome	compare certified lower-scale system against higher-scale baseline lacking the invariant package	certified system passes and matched complexity-only baseline fails or remains undefined	scale-only model passes despite missing invariants
PRED-04	‘P3-01‘	null-regime persistence under admissible dynamics	initialize near the null regime under Null-Regime Instability assumptions	stable null occupancy should fail or trajectories should leave the null regime	admissible null occupancy persists; null-stable rival survives
PRED-05	‘P2-03‘	continuity/ fragmentation pattern	compare rigid regime against finite-mass or deformable regime under matched disturbance	rigid regime keeps continuity while limited regime fragments	no differential continuity pattern appears; regime-agnostic rival survives

ID	Claim	Observable	Manipulation	Support condition	Destruction condition / rival divergence
PRED-06	‘P4-01‘	admissibility verdict under tamper and sham controls	inject hash corruption, malformed logs, or severe protocol violations	tampered runs are discarded and never promoted to evidential support	loose evaluation pipeline accepts tampered runs
PRED-07	‘P5-03‘	stability of class membership under bounded perturbation	apply perturbations inside the certified margin budget $\delta_\star^{\mathrm{cal}}(S)$	membership and geometry remain stable inside the certified budget	fragile-threshold model flips class under sub-critical perturbations
PRED-08	‘P5-01‘	non-emptiness of Q_{op} after successful certification	run the full certification schema on a system with all margins positive	Q_{op} is non-empty and operationally well-defined	certification passes while its required domain or certified equivalence is undefined; nontriviality requires a separate decoder-visible witness
PRED-09	‘P4-03‘	claim survival under latency / compute-budget stress	exceed pre-registered budget limits	strong operational claims are invalidated or downgraded	metric-first pipeline keeps claims active after budget violation
PRED-10	‘P5-05‘	legibility under noise and structured compression	raise admissible noise/compression while preserving or breaking I_{leg}	separability and intervention fidelity survive only while I_{leg} remains positive	generic observability model predicts no specific dependence on I_{leg}
PRED-11	‘P5-01‘	certification under destroyed integration and preserved complexity	destroy integration while preserving complexity or gross activity	integration-destroyed, complexity-preserved control fails certification under frozen rules	certification persists after verified integration loss; complexity-only rival survives

The matrix has two immediate consequences. First, the framework is more exposed than a purely interpretive theory because the failure thresholds are explicit. Second, not every prediction is equally mature: some now have robust internal support, while others remain provisional or blocked by implementation quality.

7 Failure Modes

7.1 What Would Count Against the Framework

The relevant failure modes are not all equally strong. Some would refute a strong reading, some would only weaken a claim, and some would reveal a technical limitation of the current program rather than of the canon itself.

Claim target	Destruction condition	Weakening condition	Current blocker class
‘P5-01‘	a system satisfies the full criterion yet fails to define the claimed operational class	boundary-family convergence would downgrade if it failed outside the present frozen internal contract	epistemic-level
‘P5-02‘	a structurally preserved pair falls into different classes under the same admissible comparison regime	one legacy raw continuous/discrete positive pair remains ambiguous on I_{ri} handling	metric/tolerance-level
‘P5-03‘	small admissible perturbations force uncontrolled class exit across stable positive cases	threshold sensitivity persists only in a narrow knife-edge zone	metric/tolerance-level
‘P5-04‘	a system loses a critical invariant and remains certified without downgrade	boundary-family convergence would downgrade if it failed outside the present frozen internal contract	epistemic-level
‘P5-05‘	a complexity-only control without preserved invariants passes certification	matched external baselines are still missing	epistemic-level
‘P5-06‘	biology is shown to be primitive inside the same operational criterion	it inherits the same localized substrate-equivalence caveat as ‘P5-02‘	metric/tolerance-level

7.2 Failure-Mode Taxonomy

Criterion 7.1 (Status meaning)**.** *Within the current scientific program, a result belongs to:*

1. `ROBUST_SUPPORT` *if the target claim survives repeated internal discrimination without an active ambiguity on the tested target;*
2. `PROVISIONAL_SUPPORT` *if the target is strengthened but still depends on a localized unresolved bottleneck;*
3. `STILL_AMBIGUOUS` *if the current tests do not cleanly separate support from failure;*
4. `IMPLEMENTATION_LIMIT` *if the current bottleneck is primarily due to probe, runner, or case-construction quality;*
5. `METRIC_OR_TOLERANCE_LIMIT` *if the bottleneck is mainly due to normalization, comparison regime, or threshold handling.*

This taxonomy is not part of the ontology of the framework. It is part of the scientific discipline around how internal evidence is reported.

7.3 Refutation Classes

For publication-level evaluation, failures should be classified by what they touch. The same failed run can be decisive, ambiguous, or merely technical depending on whether the packet was admissible and whether the manipulation was specific.

Definition 7.2 (Doctrinal Refutation Candidate)**.** *A result is a doctrinal refutation candidate for a claim family when it satisfies all admissibility gates, uses the frozen definitions, targets the declared binding coordinate, and reaches the claim's destruction condition without relying on a protocol violation.*

Definition 7.3 (Methodological Defeat)**.** *A result is a methodological defeat when the experiment is admissible but shows that the current test design, judge, threshold, comparator, or observable cannot support the intended claim. It weakens the evidence program without by itself showing the formal claim false.*

Definition 7.4 (Artifact Failure)**.** *A result is an artifact failure when the packet, manifest, runner, exclusion ledger, or analysis path fails before interpretation. Artifact failures can be important engineering evidence, but they are not positive or negative scientific support for the theoretical claim.*

Criterion 7.5 (Specificity of Failure)**.** *A failure is specific only if the failed coordinate is identifiable after controlling for gross task failure, missing observability, manifest drift, and runtime-budget violation. Non-specific failures are reported as methodological or artifact failures according to the earliest failed gate.*

Failure class	Minimum evidence required	Scientific consequence
Doctrinal refutation candidate	admissible packet set, frozen definitions, specific manipulation, destruction condition reached	claim family must be rejected, downgraded, or formally restricted
Methodological defeat	admissible packet set but non-decisive target, weak comparator, unstable threshold, or inadequate observable	evidence program must be repaired before support can be claimed
Artifact failure	packet, manifest, build, judge, or analysis trace fails before interpretation	no claim-side inference; engineering repair required
Ambiguous boundary result	admissible but falls inside a pre-declared uncertainty band	no escalation; boundary sampling or metric refinement required

8 Binding Predictions to Tests

The tables above state what would count as success or failure in broad terms. The ledger below binds the main *Six-Predicate Operational Criterion* claim families to the narrowest concrete test classes that currently matter for publication-level assessment.

Claim	Primary positive test	Primary failure test	Present evidentiary bottleneck	Current status
‘P5-01‘	successful certification with positive invariant margins and recoverable non-trivial Q_{op} geometry	certified case with no recoverable operational class, or clean existence-boundary degradation without class change	support is now cross-family robust internally, but the nearest positive member still sits close to the legibility margin	ROBUST SUPPORT W/ LOCALIZED CAVEAT
‘P5-02‘	cross-substrate pair with verified primitive correspondence and one positive criterion witness and admissible geometry match	structurally preserved pair judged non-equivalent under the same comparison regime	strongest support is robust internally but one earlier raw continuous/discrete pair remains ambiguous on I_{ri} handling	ROBUST SUPPORT W/ LOCALIZED CAVEAT
‘P5-03‘	stable class membership inside certified perturbation budget $\delta_{\star}^{cal}(S)$	class flip under sub-critical admissible perturbations across otherwise stable positive cases	one knife-edge negative-control zone remains locally fragile	PROVISIONAL SUPPORT LOCALIZED
‘P5-04‘	targeted invariant rupture causing certification exit or undefined Q_{op}	verified invariant loss without downgrade or class exit	support is now cross-family robust internally, but the evidence still lives inside the current invariant package and thresholds fixed before evaluation	ROBUST SUPPORT W/ LOCALIZED CAVEAT
‘P5-05‘	complexity-rich and connectivity-rich negative controls fail certification under stricter internal scoring protocols	matched scale-only or activity-rich controls pass despite missing invariants	no external matched-baseline campaign yet	ROBUST INTERNAL SUPPORT
‘P5-06‘	same positive equivalence path as ‘P5-02‘, interpreted without substrate privilege	biology shown to be primitive under the same operational criterion	inherits the same localized I_{ri}-handling caveat as ‘P5-02‘	ROBUST SUPPORT W/ LOCALIZED CAVEAT

9 Boundary Diagnostics

The current internal program does not fail uniformly. Its informative content lies in where the boundaries are sharp, where they are merely narrow, and where probe design still dominates the result.

Boundary target	Observed empirical profile	What it means	Limit class
Complexity-only negative control	stable fail region under stricter internal scoring, partially blinded ordering, and same-host replication across execution profiles	the criterion is not collapsing into gross activity or connectivity proxies	ROBUST INTERNAL SUPPORT
Threshold-sensitive negative region	narrow decision band rather than broad collapse	the framework has a localized edge effect, not universal threshold instability	METRIC OR TOLERANCE LIMIT
Convergent boundary-probe families	two distinct probe families now reproduce the transition pattern with localized first-fail behavior under an isolated snapshot evaluator	the existence/rupture boundary is now robust internally, though still internal-only and threshold-bound near the positive edge	ROBUST SUPPORT W/ LOCALIZED CAVEAT
Harder positive substrate families	family convergence now supports class preservation, but one earlier raw pair remains ambiguous	the live ambiguity is concentrated on how rigid-identity preservation is measured, not on every invariant at once	METRIC OR TOLERANCE LIMIT
Refined negative substrate families	clean fail across both the earlier and strengthened pair families	the equivalence criterion can reject stronger non-equivalent pairs under the fixed internal evaluator protocol and out-of-profile reproduction checks	ROBUST SUPPORT W/ LOCALIZED CAVEAT

9.1 Boundary Geometry

Boundary results are not just pass/fail outcomes. They should describe the local shape of the decision surface. Let z denote a controlled perturbation coordinate and let $Q(z)$ denote the operational certification score or verdict. Three patterns are scientifically distinct:

1. **stable interior:** $Q(z)$ remains inside the certified region for a non-trivial interval;
2. **localized transition:** $Q(z)$ crosses the decision boundary in a narrow interval tied to a declared coordinate;
3. **diffuse instability:** small unrelated perturbations cause broad and unstructured verdict changes.

Criterion 9.1 (Boundary Adequacy)**.** *A boundary probe is adequate only if it can distinguish localized transition from diffuse instability. A probe that only reports a binary verdict without the perturbation coordinate and margin behavior is insufficient for boundary diagnosis.*

Proposition 9.2 (Localized Transition Reading)**.** *If a family shows a stable interior, a localized transition, and a reproducible first-binding coordinate under a fixed evaluator, then the result supports a boundary-sensitive reading more strongly than a single binary pass/fail run.*

Proof. A binary run can be explained by threshold accident, scenario difficulty, or implementation noise. A localized transition with a reproducible first-binding coordinate adds structure: it

identifies where the decision changes and which coordinate controls the change. That additional structure constrains rival explanations that would predict arbitrary flips or no coordinate-specific transition. □

10 Minimal Reproducibility and Admissibility Discipline

The prediction layer is only credible if the evaluation path is at least partly insulated from the generator and if the evidentiary inputs are sealed before scoring. The current program does not yet have an external evaluator or an external laboratory replication, but it does enforce a stricter internal protocol than the initial baseline.

Protocol element	Current state	Residual limitation
Sealed manifests and hashes	active across post-baseline evaluations; evaluator reads sealed snapshots instead of live mutable state	still restricted to one implementation family and internal benchmark suite
Evaluator-side threshold file	active; thresholds are fixed before evaluation and logged explicitly	threshold bands still require stress analysis near knife-edge zones
Label-blinded ordering	active in partially blinded evaluations; condition labels remain hidden until after decision emission	not a full blind protocol and not an external evaluator
Multi-profile replication	active; same-host replication across distinct execution profiles preserved decisions	not external replication and not cross-laboratory agreement
Negative-control and sham policy	enforced from baseline onward and tightened in stress suites	matched external benchmark families are still missing
Evaluator isolation	isolated from live generator step functions during scoring	still not an independently maintained external codebase

This section is intended solely to distinguish internal methodological discipline from external confirmation.

10.1 Artifact Contract for Prediction Claims

Every prediction claim should be backed by a finite evidence bundle:

$$\mathcal{B} = (\mathcal{M}, \mathcal{P}, \mathcal{J}, \mathcal{R}, \mathcal{A}),$$

where $\mathcal{M}$ is the manifest set, $\mathcal{P}$ the observation packet set, $\mathcal{J}$ the executable evaluator code and threshold records, $\mathcal{R}$ the run/exclusion ledger, and $\mathcal{A}$ the executable analysis scripts. The bundle is part of the claim surface. A table without the bundle is a summary, not independently checkable evidence.

Criterion 10.1 (Bundle Sufficiency)**.** *For any status stronger than* `STILL_AMBIGUOUS`*, the evidence bundle must allow an evaluator to reconstruct the reported verdicts from sealed inputs without manual row selection.*

Criterion 10.2 (Externalization Readiness)**.** *A prediction family is ready for external evaluation only when its bundle, thresholds, exclusions, and decision rules are exportable without access to live generator state or informal operator knowledge.*

11 Current Status of Internal Support

11.1 Interpretive Discipline

The status ledger below does not reclassify the formal theorems of *Six-Predicate Operational Criterion*. Those remain formal results inside the underlying mathematical framework. The ledger classifies the *current internal scientific support* for experimentally relevant readings of those claims.

Claim	Current status	What strengthened it	What still limits it	Status scope
‘P5-01‘	ROBUST SUPPORT W/ LOCALIZED CAVEAT	near-identity boundary-probe and offset-dispersion families both reproduced PASS/ AMBIGUOUS/FAIL under the isolated snapshot evaluator and out-of-profile reproduction	support remains internal-only and the nearest positive case still sits close to the legibility margin	INTERNAL SUPPORT ONLY
‘P5-02‘	ROBUST SUPPORT W/ LOCALIZED CAVEAT	normalized-I_{ri} family and raw quantized positive/negative pair family stayed stable under the isolated evaluator and multi-profile rerun	one earlier raw continuous/discrete positive pair remains ambiguous and keeps a localized I_{ri} caveat alive	INTERNAL SUPPORT ONLY
‘P5-03‘	PROVISIONAL SUPPORT LOCALIZED	threshold-stress sweeps and boundary-isolation checks localized instability rather than showing global collapse	knife-edge threshold band persists for one negative-control region	INTERNAL SUPPORT ONLY
‘P5-04‘	ROBUST SUPPORT W/ LOCALIZED CAVEAT	two distinct boundary families now converge on rupture-side PASS/ AMBIGUOUS/FAIL behavior, with I_{ri} becoming the first binding invariant on ambiguous or failing members	the evidence remains internal-only and tied to the current invariant package plus thresholds fixed before evaluation	INTERNAL SUPPORT ONLY
‘P5-05‘	ROBUST INTERNAL SUPPORT	complexity-rich and connectivity-rich controls failed certification across stricter internal scoring protocols and label-blinded evaluation	no external matched-baseline campaign exists yet	INTERNAL SUPPORT ONLY
‘P5-06‘	ROBUST SUPPORT W/ LOCALIZED CAVEAT	same strengthened two-family substrate-equivalence path as ‘P5-02‘	it inherits the same localized I_{ri} metric-handling caveat as ‘P5-02‘	INTERNAL SUPPORT ONLY

11.2 Claim-Specific Diagnostic Notes

‘P5-05‘. This is currently the strongest experimentally discriminated claim family. The framework repeatedly rejected a complexity-rich, connectivity-rich negative control that still failed the invariant package. That result survived stricter internal scoring protocols and label-blinded evaluation. The support is therefore `ROBUST_SUPPORT`, but it remains `INTERNAL_SUPPORT_ONLY`.

‘P5-02‘ and ‘P5-06‘. These two claims move together in the current program. They were strengthened first by conservative normalized-I_{ri} handling, then by a second substrate family, a more independent evaluator path, and multi-profile confirmation. That is enough for `ROBUST_SUPPORT_WITH_LOCALIZED_CAVEAT`. The localized caveat remains real: one earlier raw continuous/discrete positive pair is still ambiguous, so the present support is robust internally but not caveat-free.

‘P5-03‘. Approximate stability is not in collapse. Threshold stress did not show universal fragility. It showed a narrow knife-edge zone around one thresholded negative-control region. The right status is therefore `PROVISIONAL_SUPPORT` with an explicit `METRIC_OR_TOLERANCE_LIMIT`.

‘P5-01‘ and ‘P5-04‘. These boundary-sensitive claims no longer rest on a single weak probe family. After the localized threshold-resolution checks and the offset-dispersion boundary control study, the refined near-identity boundary probe and the offset-dispersion family both reproduced a PASS/AMBIGUOUS/FAIL boundary pattern under the isolated snapshot evaluator and out-of-profile reproduction. That is enough for `ROBUST_SUPPORT_WITH_LOCALIZED_CAVEAT`. The caveat remains explicit: support is still internal-only, and the strongest positive member stays near the legibility margin under the fixed evaluator contract.

11.3 Methodological Status

The scientific program itself also has a status, distinct from the doctrine and distinct from the claim ledger.

Program element	Current status	Why it matters
Generator/evaluator separation	materially improved	reduces live generator-state dependence present in the initial baseline
Sealed manifests and thresholds	active	prevents live-state leakage during evaluation
Label-blinded evaluation	active but partial	reduces bias; does not equal full blind evaluation
Threshold stress testing	localized caveat	shows local knife-edge behavior instead of broad collapse
Multi-profile replication	passed internally	strengthens reproducibility discipline without constituting external replication
Isolated release re-instantiation	passed internally	sealed package was rebuilt in a clean isolated environment on the same host with no divergence during verification
Independent external evaluator codebase	absent	keeps all current support inside the internal program

Release-level reproduction note. In addition to evaluator-isolation safeguards, the sealed benchmark package was re-instantiated in a clean isolated environment on the same host without tracked divergence during verification. This strengthens internal reproducibility and handoff discipline at the verification layer. It does *not* count as external validation, third-party confirmation, or a claim upgrade by itself.

12 Cumulative Internal Evidence Map

The current status ledger is not based on a single benchmark run. It reflects a progressive sequence of internal validation campaigns. The table below records only the highest-value methodological consequence of each campaign, not every local metric.

Campaign	Main question	Strongest result	Main residual	Claim effect
Baseline protocol	can the criterion distinguish a rich negative control and survive obvious ablations?	complexity-rich negative control failed; invariant-loss ablations exited certification; proxy cross-substrate pair passed	all support remained internal and the equivalence test was still a proxy	strengthened ‘P5-05‘; gave early support to ‘P5-01‘, ‘P5-02‘, ‘P5-04‘, ‘P5-06‘
Generator separation	does the program survive stronger methodological separation and harder structural tests?	generator/evaluator separation improved; harder pair passed; near-miss family and graded curves behaved coherently	evaluator still remained internal and single-lineage	strengthened ‘P5-01‘, ‘P5-02‘, ‘P5-03‘, ‘P5-04‘, ‘P5-06‘ internally
Partially blinded stress evaluation	do results survive label-blinded evaluation, threshold stress, and multi-profile replication?	blind ordering and multi-profile replication preserved decisions; complexity-only failure remained strong	harder equivalence, near-miss boundary, and threshold stress became ambiguous	robustly strengthened ‘P5-05‘; left ‘P5-02‘, ‘P5-06‘, ‘P5-03‘ narrower
Boundary localization	can threshold fragility, near-miss ambiguity, and harder substrate ambiguity be localized?	threshold fragility was localized; refined negative pair separated cleanly; ambiguity concentrated on I_{ri} handling and near-miss probe quality	positive stronger pair remained ambiguity-bound; near-identity probe still entangled	localized rather than promoted ‘P5-02‘, ‘P5-03‘, ‘P5-04‘
Metric normalization	can probe quality and I_{ri} handling be stabilized without doctrinal weakening?	normalized-I_{ri} handling provisionally stabilized the harder positive pair; first-generation boundary probe clarified the probe weakness	earlier boundary probe remained too coarse to resolve the sharp edge	strengthened ‘P5-02‘, ‘P5-06‘; kept ‘P5-01‘, ‘P5-04‘ ambiguous
Boundary resolution	can the open boundary residuals be resolved without loosening thresholds?	refined near-identity boundary probe produced PASS/ AMBIGUOUS/FAIL with I_{ri} first-fail; refined substrate families localized ambiguity to I_{ri} handling	the results were still only localized and internal	promoted both residual families to stronger localized support
Cross-family evaluator checks	do those gains survive a more independent evaluator path and a second perturbation family?	substrate-equivalence converged across two pair families; existence/rupture converged across two probe families under isolated snapshot evaluation and out-of-profile reproduction	all support remained internal-only and a localized earlier caveat survived for ‘P5-02‘/‘P5-06‘	promoted ‘P5-01‘, ‘P5-02‘, ‘P5-04‘, ‘P5-06‘ to robust support with localized caveats

13 Priority Next Discriminators

The current state of the program already suggests which next tests have the highest diagnostic leverage. The objective is discriminating precision rather than breadth: each next test should either upgrade a provisional claim or localize an ambiguity into a measurable implementation limit.

Priority target	Why it matters now	What the next narrow result would change	Linked claims
Independent out-of-program rerun of the two boundary families	current rupture/existence support is now robust internally but still lacks outside-program confirmation	an outside reproduction would test whether the family convergence is implementation-specific or genuinely portable	‘P5-01‘, ‘P5-04‘
Independent out-of-program rerun of both substrate families	current strongest positive equivalence path is robust internally but still carries a localized earlier I_{ri} caveat	an outside reproduction would test whether the strengthened equivalence path survives beyond the internal evaluator lineage	‘P5-02‘, ‘P5-06‘
Matched external complexity baselines	the complexity-only result is internally strong but still lacks cross-program baseline pressure	a matched baseline campaign would turn the strongest present internal discriminator into an externally calibrated one	‘P5-05‘
Independent external evaluator implementation	current isolation safeguards remain within one engineering lineage	a separately maintained evaluator would reduce same-lineage measurement dependence without touching formal theory	all operational readings

14 Residual Technical Caveats and Interpretive Scopes

14.1 Residual Technical Caveats

The remaining caveats are narrow enough to state cleanly:

1. The strongest positive substrate-equivalence path is still partly dependent on normalized I_{ri} handling. This is a `METRIC_OR_TOLERANCE_LIMIT`, not a robust positive closure.

2. The earlier coarse near-identity probe remains an `IMPLEMENTATION_LIMIT`. Although cleaner than naive baselines, in the tested band it under-stresses the intended boundary.

3. Threshold stress is not globally problematic, but it is not perfectly flat either. The current result is localized rather than globally robust.

4. All empirical support remains `INTERNAL_SUPPORT_ONLY`. No external laboratory, independent external evaluator, or external benchmark suite has confirmed the framework. Isolated clean-environment re-instantiation strengthens local release reproducibility only; it does not constitute external validation.

15 Final Claim Freeze Table

The present paper specifies a prediction-and-falsation contract governed by thresholds fixed before evaluation rather than post hoc revision. The table below records the narrowest honest state for the main *Six-Predicate Operational Criterion* claim families under the current internal program.

Claim	Status	Current blocker	Next step
‘P5-01‘	ROBUST SUPPORT LOCALIZED CAVEAT	support is now strong internally but still nearest to the legibility margin on the positive edge	rerun the convergent probe families under an independently maintained external evaluator
‘P5-02‘	ROBUST SUPPORT LOCALIZED CAVEAT	one earlier raw continuous/discrete positive pair remains ambiguous on I_{ri} handling	reproduce both strengthened substrate families under an independent evaluator without post hoc threshold loosening
‘P5-03‘	PROVISIONAL SUPPORT LOCALIZED	one localized knife-edge threshold band remains active in a negative-control region	rerun the localized stress zone with denser threshold sampling and fixed evaluator contract
‘P5-04‘	ROBUST SUPPORT LOCALIZED CAVEAT	rupture-side support is strong internally but still bounded to the present invariant package and thresholds fixed before evaluation	rerun the same boundary families under an independently maintained evaluator and sealed protocol package
‘P5-05‘	ROBUST INTERNAL SUPPORT	no external matched-baseline campaign exists yet	add matched external complexity/activity baselines under the same evaluator contract
‘P5-06‘	ROBUST SUPPORT LOCALIZED CAVEAT	inherits the same localized substrate-equivalence caveat as ‘P5-02‘	upgrade only if the strengthened substrate families survive an independent external rerun

15.1 Binding Predictions to an External Biological Interface

Each proposed external falsifier must instantiate the experimental interface contract of Paper IV: population and independent unit, preparation, target/sham/disconnection/restoration operators, measured channels and units, frozen blinded decoder, calibrated error, matched information/cost, and identifying causal assumptions. A biological preparation is a candidate realization of the abstract model, not an automatic instance of its invariants. A missing realization or observation map leaves the prediction unevaluable on that preparation.

For a directional predicted margin η, preregister the score, measurement headroom and multiplicity before data. Under the independent-block contract, the lower rejection boundary is $\eta - t_n - 2Le$, with the false-rejection and power statements of Paper IV. Distinguish a valid adverse contrast from invalid sham matching, sensor dropout, leakage or an unidentified intervention. Equal recorded laws for candidate and rival make separation impossible on that interface. No simulation or terminology can remove that identification obstruction. This is a falsation specification, not an executed biological campaign; external support remains uncertified.

15.2 Constructive Indexed and Metric-Sector Discriminators

The native asymmetric infinite plant of Papers V, VII and VIII supplies a joint instrumented certificate rather than seven disconnected score flags. Its new predictions are conditional on its equations and comparison grammar: paired valuation pulses give responses 4385/262144 and 4321/262144; rate symmetrization makes their signed contrast zero. Ownership collapse, privileged-history fracture and an exact added clone destroy the corresponding conjunctive margins. The reserved three-port task has weak loss 14/15 and native loss 3/16 in its common three-operation language; an unrestricted same-cost emulator removes that advantage. These are analytic/synthetic discriminators, not measurements from a physical or biological submission.

Paper I's wedge result is likewise sector-specific: fixing positive factor diameters and bi-Lipschitz constants gives a positive Hausdorff gap and a half-gap perturbation radius. Letting a

factor shrink or the lower Lipschitz constant vanish defeats the conclusion. Its protected/damage actuation contrast uses a stated operational cost model; a real cheap clamp in the same sector would preclude the strict resistance threshold. No experimental claim may combine those two incompatible premises or transfer the radius to arbitrary homeomorphisms. Fresh external tests must identify these observation, control, energy and grammar assumptions through the Paper IV interface.

16 Conclusion

The canon now has a prediction layer and a failure-mode ledger with explicit discriminators, explicit weakening conditions, and explicit support classes.

The current internal picture is sharp enough to summarize without inflation. The complexity/connectivity insufficiency claim has robust internal support. Substrate-preserved class membership and biological non-primacy now have robust support with localized caveats. Approximate stability is provisionally supported with a localized threshold caveat. Boundary-sensitive stress of existence and rupture now has robust support with localized caveats under convergent internal probe families and a more independent judge path. All of this nevertheless remains internal-only.

This remains an internal prediction-and-falsation specification. It does not upgrade the formal framework to external validation, theorem closure beyond the upstream papers, or unconditional empirical certification.

17 Interpretation and Open Research Links

Historical alias. Predictions, Discriminators, and Failure Modes. The alias is retained for bibliographic continuity; the technical title states the demonstrated object.

Formal anchor. The memory, interaction and feedback propositions and the explicitly versioned prediction contracts above.

Proposed interpretation. Predictions connect mathematical conditional statements to a specified experimental contract. Historical support states remain attached to their original evaluator and version.

Additional obligations. Show that the operational measurement and perturbation class implement the premises of the current theorem before transferring old support to it.

Differential tests and evidence. Re-preregister changed protocols on fresh reserve data; retain the old outcomes as history rather than silently recomputing their meaning.

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Authorial Note and Methodological Provenance

The conception, theoretical direction, mathematical architecture, development, and integration of the QICN research programme were undertaken by a single independent researcher, Johnny Andrey Pérez Ramírez, over more than one year. This account of provenance is supplied by the author. Several artificial intelligence systems and formal verification tools assisted with technical drafting, proof exploration, consistency checks, and mechanized kernels under his direction. The author retains intellectual responsibility for the programme, its claims, and any remaining errors. Analytical manuscript proofs, machine-checked kernels, and internal tests have distinct scopes; independent mathematical review and empirical validation remain separate obligations. `external_support_certified=false`.